



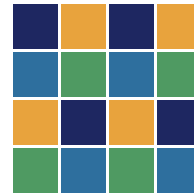
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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# Digital Image Fundamentals

*Pixel Representation, Spatial & Intensity Resolution, Grayscale Images,  
and Practical Problems in Image Quality*



**IT0512 – COMPUTER VISION**

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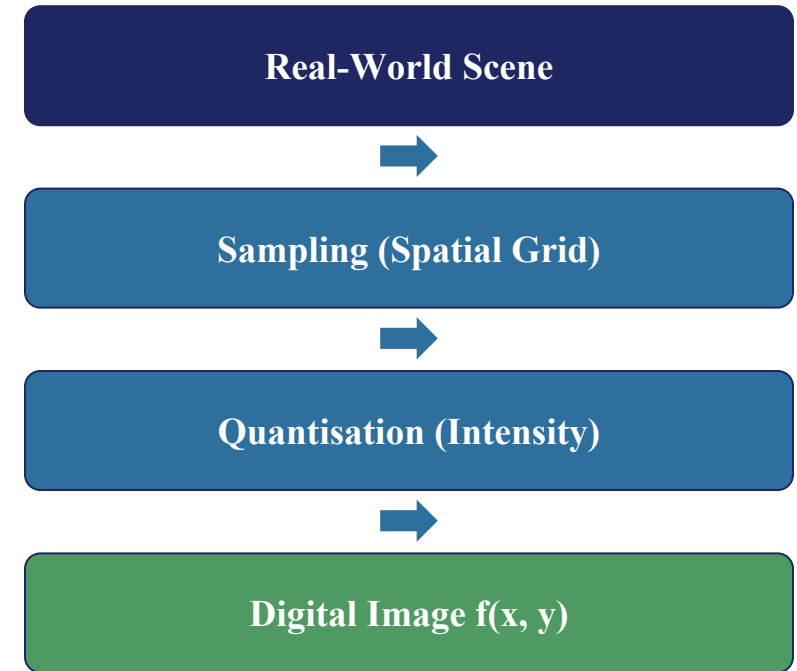
# INTRODUCTION

## What is a Digital Image?

- A digital image is a 2-D function  $f(x, y)$  sampled into a finite grid of picture elements, or pixels.
- Each pixel stores a numeric intensity (or colour) value that a computer can store, transmit, and process.
- Digitisation involves two independent steps: sampling (spatial layout) and quantisation (intensity values).

## Key Objectives

- Understand how images are represented as pixel matrices.
- Differentiate spatial resolution from intensity (gray-level) resolution.
- Study grayscale image formation and histogram analysis.
- Identify practical image-quality problems and the tools used to analyse them.



*Digitisation Pipeline*

# PIXEL REPRESENTATION

## Image as a Pixel Matrix

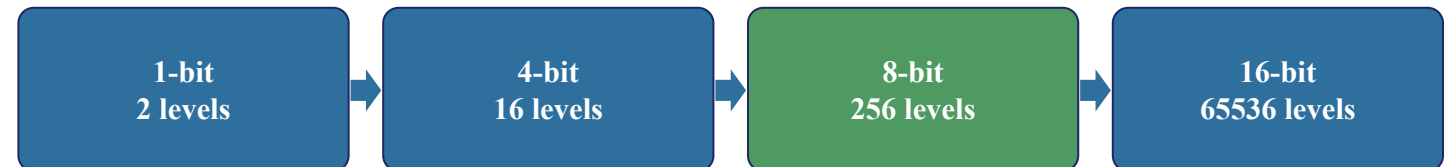
- An image of size  $M \times N$  is stored as a matrix with  $M$  rows and  $N$  columns of pixels.
- Each element  $f(x, y)$  holds the intensity at location  $(x, y)$ .
- Bit depth ( $k$ ) fixes the number of gray levels as  $2^k$  (e.g., 8-bit  $\rightarrow$  256 levels, 0-255).
- Total storage required =  $M \times N \times k$  bits.
- Colour images use three such matrices — one per Red, Green, and Blue channel.

Digital Image (512 x 512 pixels)



Zoomed 8 x 8 Pixel Grid  
(Each cell = one pixel, value 0-255)

|   |   |   |   |   |    |    |   |
|---|---|---|---|---|----|----|---|
| 5 | 4 | 5 | 6 | 5 | 5  | 6  | 5 |
| 4 | 5 | 5 | 6 | 5 | 5  | 4  | 5 |
| 6 | 5 | 5 | 6 | 6 | 4  | 6  | 5 |
| 5 | 5 | 5 | 5 | 6 | 4  | 5  | 5 |
| 4 | 5 | 5 | 5 | 5 | 5  | 5  | 5 |
| 4 | 5 | 6 | 5 | 5 | 5  | 7  | 7 |
| 4 | 5 | 6 | 5 | 6 | 8  | 14 | 8 |
| 5 | 5 | 5 | 6 | 9 | 15 | 17 | 9 |



*Bit Depth vs. Number of Gray Levels*

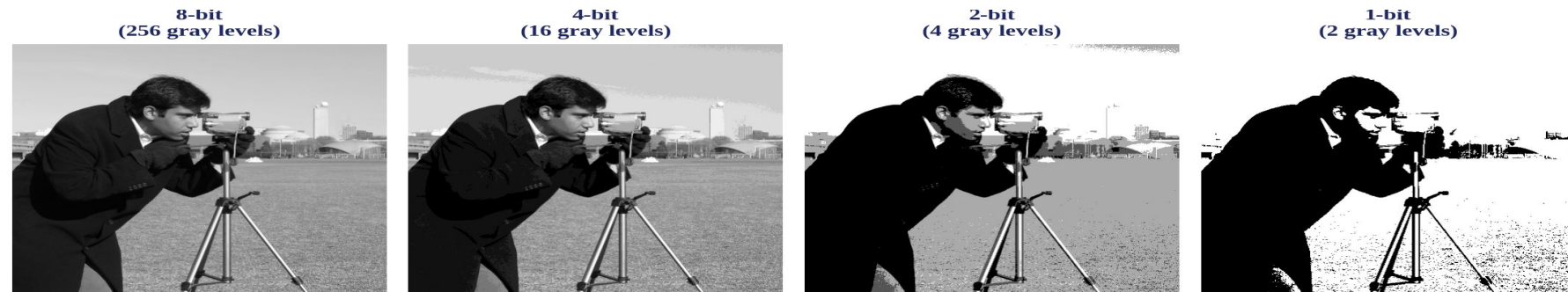
# SPATIAL & INTENSITY RESOLUTION

**Spatial Resolution** — the total number of pixels ( $M \times N$ ) used to represent an image; higher resolution preserves finer detail.

**Intensity Resolution** — the number of bits per pixel (gray levels); higher bit depth gives smoother tonal transitions.



*Effect of reducing spatial resolution (256 x 256 down to 8 x 8) — detail is progressively lost and blockiness increases.*



*Effect of reducing intensity resolution (8-bit down to 1-bit) — false contouring (banding) appears at low bit depth.*

## Definition & Conversion

- A grayscale image encodes only intensity (luminance) — no colour/hue information — using a single channel per pixel.
- Standard RGB-to-grayscale conversion (ITU-R BT.601 luminosity method):  $\text{Gray} = 0.299R + 0.587G + 0.114B$
- A histogram plots pixel count against each gray level (0-255) and reveals brightness and contrast characteristics at a glance.

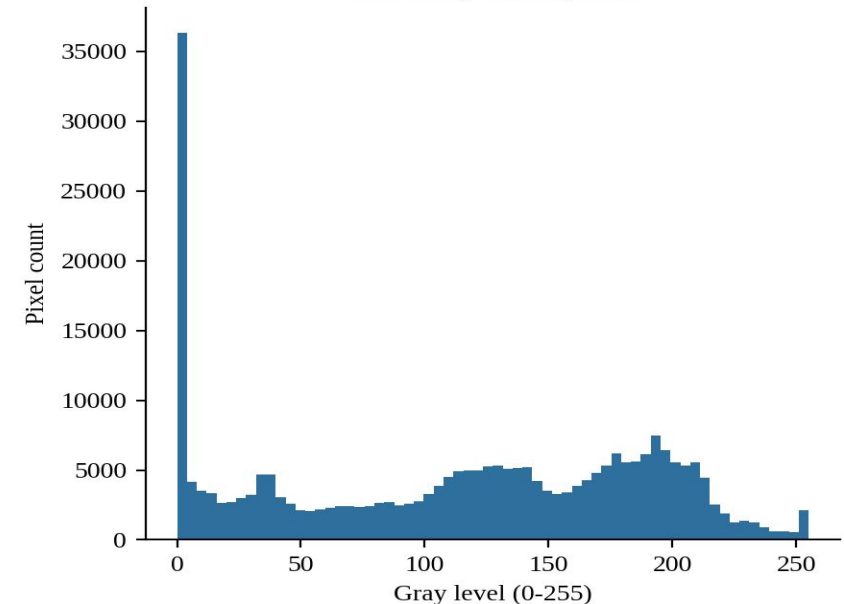
Original RGB Image



Grayscale Image  
(single intensity channel)



Intensity Histogram





# PRACTICAL PROBLEMS IN IMAGE QUALITY

## Common Quality Issues

- Noise — random intensity variations from sensor or transmission errors, degrading clarity.
- Blur — loss of sharpness from motion, defocus, or an inadequate optical system.
- Low spatial resolution — too few pixels causes visible blockiness and loss of fine detail.
- False contouring — too few gray levels (low intensity resolution) produces visible banding in smooth areas.

Original



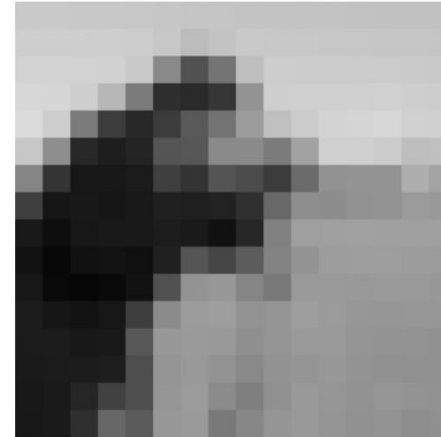
Noise



Blur



Low Spatial Resolution



False Contouring (Low Intensity Res.)



**Cause: Noise / Blur / Low Res / Low Bit-Depth**



**Effect: Weak Edges, Poor Segmentation**

**Remedy: Filtering & Enhancement**



# ANALYSIS TOOLS & CONCLUSION



## Tools Used for Analysis

- OpenCV — noise detection, filtering, edge detection, histogram equalisation.
- MATLAB — image enhancement, quantitative quality analysis.
- Python Libraries — NumPy, Pillow, Scikit-image, Matplotlib for pixel-level processing and visualisation.

## Conclusion

- Every digital image is fundamentally a matrix of quantised intensity values.
- Spatial and intensity resolution jointly determine perceptual image quality.
- Grayscale representation and histograms are foundational tools for quality analysis.
- Recognising noise, blur, and resolution limits is essential before any higher-level processing.



THANK YOU  
for your attention

